Week 3: Confounding, Selection Bias, and Colliders

Davood Tofighi, Ph.D.

Motivation and Learning Objectives

In our last module, we learned the "grammar" of causal inference using Directed Acyclic Graphs (DAGs). Now, we use this grammar to diagnose and understand the primary threats that prevent us from making valid causal claims from observational data: **confounding** and **selection bias**.

These concepts are the reasons why well-intentioned analyses can lead to dangerously wrong conclusions. For decades, observational studies suggested that hormone replacement therapy (HRT) protected against heart disease, a conclusion that was later overturned by randomized trials showing it could be harmful. What went wrong? The answer lies in confounding—the women who opted to take HRT were already healthier and had more health-conscious behaviors than those who did not.

By the end of this module, you will be able to:

1. **Define** confounding and selection bias using both the potential outcomes and causal graph frameworks.
2. **Identify** confounding (open backdoor paths) and selection bias (often caused by conditioning on a collider) in a DAG.
3. **Implement** control strategies like stratification and regression to address confounding.
4. **Recognize** subtle forms of bias, such as M-bias, and understand why "controlling for everything" can be harmful.
5. **Apply** these concepts to critically evaluate the validity of causal claims from observational studies.

Key Concepts: The Villains of Observational Research

1. Confounding: The Hidden Common Cause

Confounding is the most famous threat to causal inference. It occurs when a third variable is associated with both the exposure and the outcome, creating a spurious (non-causal) link between them.

Formal Definition (DAGs): Confounding occurs when there is an unblocked **backdoor path** between an exposure `A` and an outcome `Y`. This path is created by a **common cause** (`C`) of `A` and `Y`, represented by the "fork" structure: `A ← C → Y`.

Intuition: Imagine you observe that cities with more ice cream sales (`A`) also have more drowning deaths (`Y`). Does ice cream cause drowning? No. A third variable, hot weather (`C`), is a common cause of both. Hot weather increases ice cream sales and also causes more people to swim (and thus, more drownings). Unless we account for the weather, we will draw a wrong conclusion.

Control Strategies: To eliminate confounding, we must **block the backdoor path**. The two most common methods are:

1. **Stratification (or Restriction):** We analyze the relationship between `A` and `Y` *within levels* of the confounder `C`. For example, we could compare ice cream sales and drownings only on days with similar temperatures.
2. **Regression Adjustment:** We include the confounder `C` as a covariate in a regression model (e.g., `lm(Y ~ A + C)`). This statistically "holds `C` constant."

ℹ □ Numerical Example 1 (Stratification): The Hypertension Drug

Let's examine the effect of a new hypertension drug (`A`) on 1-year mortality (`Y`), where disease severity (`Z`) is a confounder. Sicker patients are both more likely to receive the new drug and more likely to die.
Crude (Biased) Data:

	Died (Y=1)	Survived (Y=0)	Total	Crude Mortality
New Drug (A=1)	40	60	100	**40%**
Old Drug (A=0)	20	80	100	**20%**

The **crude risk ratio** is `(40/100) / (20/100) = 2.0`. It appears the new drug *doubles* the risk of death!

Step 1: Stratify by Severity (`Z`)

High Severity (Z=1) *Stratum-specific RR = (32/80) / (10/20) = 0.4 / 0.5 = 0.8*

Low Severity (Z=0) *Stratum-specific RR = (8/20) / (10/80) = 0.4 / 0.125 = 3.2.* Let's make this example consistent.

Let's use the corrected example from the previous turn to show confounding reversal.

High Severity (Z=1)

	Died (Y=1)	Survived (Y=0)	Total	Mortality
New Drug (A=1)	32	48	80	**40%**
Old Drug (A=0)	10	10	20	**50%**

Stratum-specific RR = 0.40 / 0.50 = 0.8

Low Severity (Z=0)

	Died (Y=1)	Survived (Y=0)	Total	Mortality
New Drug (A=1)	2	18	20	**10%**
Old Drug (A=0)	10	70	80	**12.5%**

Stratum-specific RR = 0.10 / 0.125 = 0.8

Step 2: Interpretation Within each severity stratum, the new drug is **protective** (Risk Ratio = 0.8). The crude RR of 2.0 was severely biased because doctors gave the new drug to the sickest patients. Confounding masked the true effect and reversed its sign.

i □ Numerical Example 2 (Regression Adjustment)

Let's simulate a continuous example in R.

Scenario: We want to estimate the effect of daily minutes of exercise (`A`) on a continuous health score (`Y`), but this is confounded by age (`Z`).

```r
set.seed(123)
n <- 500
age <- runif(n, 20, 70) # Z: Age is a confounder
# A: Exercise decreases with age
exercise <- 120 - 1 * age + rnorm(n, 0, 20)
# Y: Health score decreases with age but increases with exercise (true effect =
0.5)
health_score <- 80 - 0.5 * age + 0.5 * exercise + rnorm(n, 0, 10)

df <- data.frame(health_score, exercise, age)

# Naive (Biased) Model: Ignores the confounder age
naive_model <- lm(health_score ~ exercise, data = df)
summary(naive_model)
#>                 Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   56.1268     1.3735   40.86   <2e-16 ***
#> exercise       0.0152     0.0189    0.80   0.422

# Adjusted (Unbiased) Model: Controls for age
adjusted_model <- lm(health_score ~ exercise + age, data = df)
summary(adjusted_model)
#>                 Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   79.6209     2.5020   31.82   <2e-16 ***
#> exercise       0.5147     0.0125   41.24   <2e-16 ***
#> age           -0.4950     0.0255  -19.41   <2e-16 ***
```

Interpretation: The naive model finds a near-zero, insignificant effect of exercise. This is because younger people exercise more and are healthier, while older people exercise less and are less healthy. The positive effect of exercise is cancelled out by the negative effect of age. Once we control for age in the adjusted model, we recover the true causal effect: each additional minute of exercise increases the health score by **0.51 points**.

2. Selection Bias and the Menace of Colliders

Selection bias is a systematic error that occurs when the process of selecting subjects into a study is related to both the exposure and the outcome.

Formal Definition (DAGs): A very common and insidious form of selection bias is **collider bias**. This occurs when we **condition on a collider**.

Intuition: Remember, a collider is a variable that is a common *effect*. Conditioning on it creates a spurious association between its causes. This act of conditioning can be explicit (adding the collider to a regression) or implicit (designing a study where the sample is defined by the collider).

> **i** □ Example: The "Low Birthweight Paradox"
>
> A famous real-world example of collider bias involves smoking and infant mortality.
>
> - **Exposure `A`**: Maternal Smoking
> - **Outcome `Y`**: Infant Mortality
> - **Unobserved Factor `U`**: Congenital malformations.
> - **Collider `C`**: Low Birthweight (LBW)
>
> **The DAG:** Smoking (`A`) can cause LBW (`C`). Certain congenital defects (`U`) also cause LBW (`C`) and strongly increase the risk of infant mortality (`Y`). The simplified DAG looks like this: `A → C ← U → Y`.
> **The Paradox:** In the general population, smoking increases infant mortality. However, if researchers restrict their analysis to *only low-birthweight babies*, they often find that infants of smoking mothers have a *lower* mortality rate than infants of non-smoking mothers.
> **The Explanation:** This is classic collider bias. We are conditioning on the collider, LBW (`C`). Among LBW babies, a child of a non-smoker *must* have another powerful reason for their low birthweight, such as a severe congenital defect (`U`). A child of a smoker might have LBW simply *because* their mother smoked. Therefore, within the LBW stratum, maternal smoking becomes spuriously associated with a lower risk of having other, more fatal, problems.

3. M-Bias: The Deceptive Confounder

Controlling for variables that occur *before* the exposure seems safe, right? Not always. **M-bias** is a specific form of collider bias that can occur when we condition on a pre-exposure variable that is *not* a true confounder.

The DAG Structure: Imagine the following structure, which looks like the letter 'M'. `A ← U1 → Z ← U2 → Y`

- `A` is the exposure, `Y` is the outcome.
- `Z` is a pre-exposure covariate we are tempted to control for.
- `U1` and `U2` are unobserved common causes.

The Problem: In this DAG, there is **no open backdoor path** between `A` and `Y`. The path `A ← U1 → Z ← U2 → Y` is naturally blocked by the collider, `Z`. The naive association between `A` and `Y` is unbiased.

However, if we **condition on `Z`**, we unblock the path by opening the collider. This creates a spurious association between `A` and `Y`, inducing bias.

> ⚠ The Pitfall of "Kitchen Sink" Regressions
>
> M-bias is a stark warning against the practice of including every available pre-exposure variable in a regression model. A variable is not a confounder just because it is associated with both the exposure and the outcome. It must be a **common cause**. You must use your substantive knowledge, encoded in a DAG, to decide which variables to adjust for.

Check Your Understanding

You are studying the effect of a new hypertension medication (`A`) on stroke risk (`Y`). You build the following DAG based on your knowledge:

`A ← Doctor's Preference (U1) → Patient's Zip Code (Z) ← Socioeconomic Status (U2) → Y`

A colleague suggests you should control for the patient's zip code (`Z`) because it's associated with both the medication prescribed and stroke risk. What do you say?

##

Answer

You Should Advise against Controlling for Zip Code. In This DAG, Zip Code (`Z`) is a Collider. It is Caused by the Doctor's Preferred Hospital Network (`U1`, Which Influences Prescribing `A`) and the Patient's Socioeconomic Status (`U2`, Which Influences Stroke risk `Y`). Controlling for `Z` Would Induce M-bias, Creating a Spurious Association between the Medication and Stroke risk where None Might Exist.

Appendix: Advanced Statistical Derivations and Theory

Motivation: Formalizing the Backdoor Adjustment

In the main text, we used intuition and numerical examples to show that conditioning on confounders gives us the causal effect. The mathematical justification for this comes from the **Backdoor Adjustment Formula**, which connects the interventional `do()`-calculus to standard conditional probabilities.

The Goal: We want to estimate $P(Y = y|do(A = a))$. We can't observe this directly. The backdoor adjustment formula shows how to compute it from observational data, provided we have measured a set of variables `Z` that satisfy the backdoor criterion.

Derivation: The Backdoor Adjustment Formula

The formula is:

$$P(Y = y|do(A = a)) = \sum_{z} P(Y = y|A = a, Z = z) \times P(Z = z)$$

Let's walk through a conceptual derivation:

1. **Start with the Goal**: We want to know the outcome distribution in a population where everyone has been forced to take exposure level `a`.
2. **Stratify the Population**: We can think of the overall population as being composed of different strata, defined by the levels of our confounders `Z`. The overall effect is the average of the effects within these strata, weighted by the proportion of the population in each stratum, $P(Z = z)$.
3. **Estimate the Effect within a Stratum**: Within a specific stratum `Z=z`, all backdoor paths are blocked. This means that the statistical association is equal to the causal effect. So, for the subgroup where `Z=z`, the outcome distribution under the intervention `do(A=a)` is just the observed conditional probability $P(Y = y|A = a, Z = z)$.
   - *Annotation*: This is the key step. Because `Z` satisfies the backdoor criterion, the causal link `A → Y` is the only thing connecting them *within a level of* Z. So, `P(Y|do(A), Z) = P(Y|A, Z)`.
4. **Average Across Strata**: To get the effect for the whole population, we average the stratum-specific effects, weighting each by that stratum's prevalence, $P(Z = z)$. This gives us the formula above.

This formula is the mathematical engine behind control strategies like stratification and regression adjustment. It formally shows how conditioning on a valid set of confounders allows us to estimate a causal effect from observational data.